

CliniScan

Milestone 1 — Data Description & Preprocessing

Project	CliniScan — Chest X-ray Abnormality Detection
Dataset	VinBigData Chest X-ray Abnormalities Detection (Kaggle)
Milestone	1 — Data Understanding & Preprocessing Pipeline
Scope	Dataset Description + Preprocessing Theory & Code

PART A — Dataset Description

1. Introduction to the Dataset

The VinBigData Chest X-ray Abnormalities Detection dataset is a large-scale, radiologist-annotated medical imaging dataset designed for automated detection of thoracic pathologies. It was published by Vingroup Big Data Institute and hosted on Kaggle as a competition dataset. Each image is a chest radiograph (CXR) stored in DICOM format, annotated with bounding boxes for up to 14 distinct disease categories by three independent board-certified radiologists.

Chest X-ray analysis is one of the most common diagnostic procedures in clinical medicine. Automating the detection and localization of abnormalities has the potential to significantly reduce radiologist workload, improve diagnostic consistency, and enable scalable screening programs in resource-constrained settings. This dataset forms the foundation of the CliniScan project, which aims to build a robust deep learning pipeline for chest X-ray abnormality detection and localization.

2. Dataset Composition

The dataset is divided into a training set and a test set. The training set consists of 15,000 chest radiographs, each accompanied by annotations from three radiologists. The test set contains 3,000 images without publicly available annotations (used for Kaggle leaderboard evaluation). For the purposes of this milestone, 50 training images were selected for pipeline development and validation.

Component	Details
Training Images	15,000 DICOM files in /train folder
Test Images	3,000 DICOM files in /test folder
Annotation File	train.csv — bounding boxes per radiologist per image
Submission File	sample_submission.csv — output format for Kaggle
Development Subset	50 images (used in this milestone)

3. DICOM Format and Image Characteristics

Images are stored in Digital Imaging and Communications in Medicine (DICOM) format — the universal standard for medical imaging data. DICOM files encapsulate not only the pixel data but also rich metadata including patient information, acquisition parameters, and imaging modality. For this project, only the pixel array is extracted and converted to PNG format for downstream processing.

A critical characteristic of DICOM chest X-rays is the PhotometricInterpretation field, which determines how pixel values map to brightness. Images tagged MONOCHROME2 follow the conventional mapping (higher value = brighter), while MONOCHROME1 images are inverted (higher value = darker). Failing to handle this distinction results in visually inverted X-rays that can mislead model training. The DICOM-to-PNG conversion step in this project explicitly checks and corrects for MONOCHROME1 images.

4. Annotation Structure

The train.csv file contains bounding box annotations in absolute pixel coordinates relative to the original DICOM image dimensions. Since three radiologists independently annotated each image, a single image may have multiple annotation rows for the same class — reflecting inter-annotator variation. The columns in train.csv are as follows:

Column	Type	Description
image_id	Categorical	Unique ID matching the DICOM filename
class_name	Categorical	Name of the pathological finding
class_id	Integer	Numeric class label (0–14)
rad_id	Categorical	Radiologist identifier (R1, R2, R3)
x_min	Float	Left edge of bounding box (pixels)
y_min	Float	Top edge of bounding box (pixels)
x_max	Float	Right edge of bounding box (pixels)
y_max	Float	Bottom edge of bounding box (pixels)

5. The 14 Pathological Classes

The dataset covers 14 clinically relevant thoracic pathologies identifiable on chest X-rays. A 15th label — 'No finding' — is used for normal scans and carries no bounding box coordinates. Images labeled 'No finding' are represented by empty label files in the YOLO annotation pipeline.

ID	Class Name	Clinical Significance
0	Aortic enlargement	Widening of aortic silhouette; may indicate hypertension or aortic disease
1	Atelectasis	Partial or complete lung collapse; impairs gas exchange
2	Calcification	Calcium deposits in lung tissue, vessels, or lymph nodes
3	Cardiomegaly	Enlarged cardiac shadow; associated with heart failure
4	Consolidation	Alveolar filling with fluid; common in pneumonia

5	ILD	Interstitial lung disease; diffuse fibrotic or inflammatory changes
6	Infiltration	Hazy opacities suggesting infection or inflammation
7	Lung Opacity	Increased density in lung fields; broad radiological finding
8	Nodule/Mass	Discrete round lesions; may indicate malignancy
9	Other lesion	Miscellaneous findings not covered by specific classes
10	Pleural effusion	Fluid in pleural space; often seen with heart failure or infection
11	Pleural thickening	Fibrous thickening of pleural lining; may follow inflammation
12	Pneumothorax	Air in pleural cavity causing partial lung collapse
13	Pulmonary fibrosis	Progressive scarring of lung tissue; reduces lung compliance
14	No finding	Normal scan — no pathology identified

6. Annotation Conversion — CSV to YOLO Format

Raw annotations in train.csv use absolute pixel coordinates in the original DICOM resolution. To prepare annotations for YOLOv8 training, they are converted to YOLO format, which requires coordinates normalized to [0, 1] relative to the image dimensions. The conversion formula is:

```
center_x = ((x_min + x_max) / 2) / image_width
```

```
center_y = ((y_min + y_max) / 2) / image_height
```

```
width = (x_max - x_min) / image_width
```

```
height = (y_max - y_min) / image_height
```

Since images are resized from their original DICOM resolution to 512×512 during preprocessing, bounding box coordinates are first scaled by a factor of 0.5 (512 / 1024) before normalization. Each image receives a corresponding .txt file with one YOLO annotation line per bounding box. Images with 'No finding' receive empty .txt files, as required by the YOLO training format. A dataset.yaml file is also generated to define class names and dataset paths for YOLOv8.

PART B — Data Preprocessing Theory

1. Overview of the Preprocessing Pipeline

Medical image preprocessing is a critical step that directly impacts model performance. Raw chest X-rays from DICOM files contain significant variability in resolution, pixel intensity distribution, contrast, and noise — all of which can hinder a neural network's ability to learn meaningful features. The preprocessing pipeline implemented in this project applies a series of deterministic transformations to standardize images before they are fed into any detection model.

The pipeline is designed around two key principles: (1) preserve clinically relevant information such as fine lung textures, vessel patterns, and lesion boundaries; and (2) remove or reduce sources of variability that are irrelevant to pathology detection, such as scanner-specific intensity scaling, border noise, and image-level contrast differences. The steps are applied in a fixed sequence to every image.

2. DICOM to PNG Conversion

The first step converts raw DICOM files to PNG format using the pydicom library. The pixel array is extracted directly from the DICOM object, bypassing all DICOM rendering pipelines. This ensures consistent access to the raw sensor values. As noted in the dataset description, images with PhotometricInterpretation = MONOCHROME1 are inverted before saving, so that all output PNGs follow the same convention (brighter = higher pixel value). The pixel values are then normalized to the 0–255 range using OpenCV's NORM_MINMAX normalization and saved as 8-bit unsigned integer PNGs.

Code used for DICOM conversion:

```
ds = pydicom.dcmread(dicom_path)
img = ds.pixel_array
if ds.PhotometricInterpretation == 'MONOCHROME1': img = np.max(img) - img
img = cv2.normalize(img, None, 0, 255, cv2.NORM_MINMAX)
img = np.uint8(img)
```

3. Grayscale Loading

Although the images are already grayscale chest X-rays, OpenCV's default imread loads images as 3-channel BGR. To avoid unnecessary computation and memory usage in downstream steps, images are loaded directly in grayscale mode using the IMREAD_GRAYSCALE flag. This produces a single-channel 2D numpy array, which is the correct format for all subsequent processing steps and for direct input to grayscale-aware neural network pipelines.

```
img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
```

4. Border Cropping

Chest X-ray images frequently contain non-diagnostic border regions: black padding, scanner edge artifacts, and background areas that carry no pathological information. Retaining these regions wastes model capacity and can introduce spurious patterns. A 2% crop is applied symmetrically to all four edges of the image, removing these border artifacts while retaining the full lung field and surrounding anatomy.

```
h, w = img.shape
img = img[int(h*0.02) : int(h*0.98), int(w*0.02) : int(w*0.98)]
```

5. Resizing

Original VinBigData DICOM images vary significantly in resolution, with some exceeding 3000×4000 pixels. Training a deep learning model on variable-resolution images is not feasible, as all images in a batch must share the same spatial dimensions. All images are resized to a fixed 512×512 pixel resolution using the INTER_AREA interpolation method.

INTER_AREA is the preferred choice for downsampling because it computes the weighted average of pixels in each target cell's source region, producing smoother results and avoiding the aliasing artifacts that affect INTER_LINEAR and INTER_NEAREST for large downscale factors. 512×512 is chosen as a balance between retaining sufficient anatomical detail and keeping computational cost tractable for training.

```
img = cv2.resize(img, (512, 512), interpolation=cv2.INTER_AREA)
```

6. Intensity Normalization — Min-Max

Different scanners and acquisition protocols produce images with widely varying pixel intensity ranges. A scanner may produce 12-bit or 16-bit DICOM pixel values, and even after initial 0–255 mapping during DICOM conversion, residual intensity variation exists across images. Min-Max normalization re-scales each image so that its minimum pixel value maps to 0 and its maximum maps to 255.

This step is performed using OpenCV's `cv2.normalize` with `NORM_MINMAX`, which is consistent with the approach used in the DICOM conversion step. The result is that every image utilizes the full 8-bit intensity range, improving the signal available for contrast enhancement in subsequent steps and ensuring consistent input statistics for the neural network.

```
img = cv2.normalize(img, None, 0, 255, cv2.NORM_MINMAX)
img = np.uint8(img)
```

7. CLAHE — Contrast Limited Adaptive Histogram Equalization

Chest X-rays often exhibit low local contrast in regions of clinical interest — for example, subtle nodules embedded in dense lung tissue, or early-stage consolidations. Global histogram equalization would enhance overall contrast but tends to over-amplify noise in homogeneous regions. CLAHE addresses this by dividing the image into small non-overlapping tiles (8×8 in this implementation) and applying histogram equalization locally within each tile.

A clip limit of 2.0 is applied to prevent excessive amplification of local contrast — any histogram bin that would exceed this threshold is clipped and the excess is redistributed uniformly. The result is enhanced visibility of fine structures such as vessel walls, interstitial patterns, and lesion boundaries, which are critical

for accurate bounding box localization.

```
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
img = clahe.apply(img)
```

8. Denoising — Gaussian Blur

Medical images contain inherent quantum noise from X-ray photon detection, as well as high-frequency noise introduced by scanner electronics. This noise can interfere with feature extraction in convolutional neural networks by creating spurious gradients in homogeneous regions. A light Gaussian blur with a 3×3 kernel is applied to suppress high-frequency noise while preserving the edges and structures that are diagnostically relevant.

A 3×3 kernel is deliberately chosen to be minimal — aggressive blurring would smooth out fine anatomical detail such as vascular markings and nodule boundaries that are essential for accurate detection. The Gaussian weighting ensures that the central pixel retains more influence than its neighbors, producing a gentler smoothing effect than a uniform mean filter.

```
img = cv2.GaussianBlur(img, (3, 3), 0)
```

9. Padding to Uniform Shape

Although the resize step targets 512×512, the border crop step may produce images with slightly non-square dimensions if the original image was not square. To guarantee every output image is exactly 512×512, zero-padding (black padding) is applied symmetrically to any remaining dimension deficit. This is computed independently for height and width, with the deficit split evenly between opposite sides.

Zero-padding is preferred over stretching or further resizing because it preserves the aspect ratio and pixel values of the existing content without introducing any additional resampling artifacts. For chest X-rays, maintaining anatomical proportions is important for accurate localization of findings.

```
img = cv2.copyMakeBorder(img, pad_top, pad_bottom, pad_left, pad_right,
                          cv2.BORDER_CONSTANT, value=0)
```

10. Summary of the Complete Pipeline

Step	Operation	Tool / Method	Purpose
1	DICOM → PNG	pydicom + cv2.normalize	Extract pixel data, fix inversion
2	Grayscale Load	cv2.IMREAD_GRAYSCALE	Single-channel, memory efficient
3	Border Crop (2%)	Array slicing	Remove non-diagnostic borders
4	Resize to 512×512	cv2.INTER_AREA	Uniform resolution for batching
5	Min-Max Normalize	cv2.NORM_MINMAX	Consistent intensity range
6	CLAHE	cv2.createCLAHE(2.0, 8×8)	Local contrast enhancement
7	Gaussian Blur	cv2.GaussianBlur (3×3)	Light noise suppression
8	Padding	cv2.copyMakeBorder	Ensure exact 512×512 output
9	YOLO Label Creation	pandas + coordinate math	Scale & normalize bbox coords

11. Why This Order Matters

The sequence of preprocessing steps is not arbitrary. Border cropping is performed before resizing so that the resize operation is not wasted on non-diagnostic border pixels. Normalization is applied before CLAHE because CLAHE operates on the histogram of the image — if the intensity range is compressed or skewed, CLAHE will produce suboptimal tile equalization. Denoising is applied after CLAHE because CLAHE can amplify local noise in smooth regions; the subsequent Gaussian blur suppresses this noise without undoing the contrast gains in structured regions. Padding is the final step to guarantee exact output dimensions regardless of intermediate rounding.